Lecture 8: Blackboard Notes (Active Nematics onward)

A suggested sequence of boards. Each block lists what to write; arrows show the logical flow. Sketches to draw are noted in italics.

Opening: why active nematics? (motivation before Board 1)

Active matter made of elongated, head-tail-symmetric units – cytoskeletal filaments driven by molecular motors, dense bacterial suspensions, sheets of elongated cells – behaves like a liquid crystal that is continuously fed energy. Guiding question: *what happens to an ordered liquid crystal when it is never allowed to reach equilibrium?*

Why each piece matters. The Q-tensor is the right language because the order is apolar ($\mathbf{n} \equiv -\mathbf{n}$), so a single velocity vector will not do. Topological defects are not blemishes here but the carriers of the dynamics. And one active term, $\sigma^{\text{active}} = -\zeta\mathbf{Q}$, is enough to turn a quiescent liquid crystal into a self-stirring one.

Surprises compared to equilibrium.

- An equilibrium nematic relaxes to a uniform, motionless ordered state; an active nematic **never settles** and flows on its own.
- Uniform order is **generically unstable**: any activity above a tiny threshold $\zeta_c \sim K/L^2$ destroys the alignment – the opposite of equilibrium, where order is stable and protected.
- Topological defects, static and slowly annihilating in equilibrium, become **self-propelled particles** that sustain "active turbulence" at zero Reynolds number.

Board 1: Liquid crystals and the isotropic-nematic transition

- **isotropic**: rods point randomly (an ordinary liquid)
- **nematic**: rods aligned along a common axis = director $\mathbf{n}$
- transition driven by **density**, not by attractive energy
- entropic "order from disorder" (Onsager):

$$F = F_{\text{ideal}} + F_{\text{excl}} = k_B T \int f \ln f \, d\Omega \; + \; \tfrac{1}{2}\rho \iint f\, f\, V_{\text{excl}}\, d\Omega_1 d\Omega_2$$

- $f(\Omega)$ = **orientational distribution function**: probability density that a rod points along the angles Ω. Uniform f = isotropic; peaked f = nematic
- first term (F_{ideal}) = orientational entropy, smallest for uniform f $\Rightarrow$ **penalizes** alignment; second term (F_{excl}) = free-volume cost, **reduced** by alignment
- excluded volume of two rods (length L, diameter D): $V_{\text{excl}} \approx 2L^2 D\,|\sin\gamma|$ (minimal when parallel)
- high ρ: free-volume gain from aligning beats orientational-entropy loss $\Rightarrow$ nematic

Sketch: a box of randomly oriented rods $\rightarrow$ aligned rods.

Board 2: Director and the Q-tensor

- polar (Vicsek) vs nematic: head-tail symmetry $\mathbf{n} \equiv -\mathbf{n}$

- **why not a vector?** with $\mathbf{n} \equiv -\mathbf{n}$ the average $\langle \mathbf{n} \rangle = 0$ even in a perfectly ordered state, so a vector cannot describe nematic order. The simplest object invariant under $\mathbf{n} \to -\mathbf{n}$ is the dyad $\mathbf{nn}$ (a rank-2 tensor)
- scalar order parameter (θ = angle to $\mathbf{n}$):

$$S = \langle 2\cos^2\theta - 1 \rangle \text{ (2D)}, \qquad S = \left\langle \tfrac{3\cos^2\theta - 1}{2} \right\rangle \text{ (3D)}, \qquad S \in [0, 1]$$

- **Q-tensor** – a microscopic average over the rods' unit orientations $\mathbf{u}$:

$$Q_{ij} = \left\langle u_i u_j - \tfrac{1}{d}\delta_{ij} \right\rangle = S\left(n_i n_j - \tfrac{1}{d}\delta_{ij}\right)$$

- properties: **symmetric** and **traceless** (the $-\frac{1}{d}\mathbf{I}$ subtracts the isotropic part, so $\mathbf{Q} = 0$ in the isotropic phase)
- explicit 2D form (director angle φ):

$$\mathbf{Q} = \frac{S}{2}\begin{pmatrix} \cos 2\varphi & \sin 2\varphi \\ \sin 2\varphi & -\cos 2\varphi \end{pmatrix}$$

note the 2φ: rotating the director by π leaves $\mathbf{Q}$ unchanged (a "spin-2" object)

- **reading n and S back out**: diagonalize $\mathbf{Q} \Rightarrow$ largest eigenvalue $\to S$, its eigenvector $\to \mathbf{n}$
- eigenvalues: 2D $\pm S/2$; 3D uniaxial $\frac{2S}{3}, -\frac{S}{3}, -\frac{S}{3}$

Board 3: Topological defects

Sketch: $+1/2$ (comet), $-1/2$ (trefoil), $+1$ (aster).

- charge $m = \dfrac{1}{2\pi}\oint d\theta$; the director turns by $2\pi m$ around the core
- Frank elastic energy (splay, twist, bend):

$$F_d = \tfrac{1}{2}K_1(\nabla\cdot\mathbf{n})^2 + \tfrac{1}{2}K_2(\mathbf{n}\cdot\nabla\times\mathbf{n})^2 + \tfrac{1}{2}K_3\big(\mathbf{n}\times(\nabla\times\mathbf{n})\big)^2$$

- defect energy $E \approx \pi K m^2 \ln(R/a)$
  - a = defect **core radius** (inner cutoff, where the elastic description breaks down); R = **system size** (outer cutoff, or the spacing to the nearest defect)
  - the ln comes from integrating the elastic energy density $\sim Km^2/r^2$ from a to R: $\int_a^R \frac{Km^2}{r^2}\, 2\pi r\, dr \sim Km^2 \ln(R/a)$
- one $+1$ ($E \propto 1$) splits into two $+1/2$ ($E \propto 2 \times \frac{1}{4} = \frac{1}{2}$): **splitting lowers the energy**
  - the **elastic free energy** drops; the released energy is dissipated **viscously as heat** as the field relaxes – total energy is conserved, entropy increases (like a ball rolling downhill)
  - **topological charge is conserved**: $+1 = +\frac{1}{2} + \frac{1}{2}$; the two like-sign halves **repel** $\Rightarrow$ they fly apart

Board 4: Active nematics – the active stress

- inject energy $\Rightarrow$ add the active stress

$$\sigma^{\text{active}} = -\zeta\,\mathbf{Q}$$

- **what a stress is** – think of it as a generalized pressure. Pressure = force per area pushing straight onto a surface; a stress is the force per area transmitted across *any* internal surface, and it can push, pull, or shear (slide)

- **imaginary-cut picture**: slice the material with an imaginary plane – the material on one side exerts a force per area on the other side across the cut, and *that* is the stress. (It depends on how the cut is oriented, which is why it is a tensor: $t_i = \sigma_{ij}\hat{m}_j$ for a cut with normal $\hat{\mathbf{m}}$.)
- a **uniform** stress pushes equally on opposite faces of a fluid blob ⇒ no net force; only a stress that **varies in space** leaves a net push, the force per volume $\mathbf{f} = \nabla \cdot \sigma$ – this is what drives the flow
- **why activity makes a stress**: each active unit pushes the fluid one way and pushes back the other way – a pair of opposite forces = a **force dipole**. Smear many over a region and they act like an internal pressure that varies in space and stirs the fluid (strength per area $\sim$ density $\times$ force $\times$ separation ⇒ units of stress)
- $\zeta > 0$ **extensile** (dipoles push fluid out along $\mathbf{n}$); $\zeta < 0$ **contractile** (pull in along $\mathbf{n}$)

Sketch: an imaginary cut with force arrows across it; a force dipole = two opposite arrows.

- momentum balance, zero Reynolds number (inertia negligible ⇒ forces balance instantly): $\nabla \cdot (\sigma^{\text{passive}} + \sigma^{\text{active}}) = \nabla p$
  - p = the (hydrodynamic) **pressure**, fixed by **incompressibility** $\nabla \cdot \mathbf{u} = 0$; it adjusts locally so the fluid neither compresses nor expands (a Lagrange multiplier for that constraint)
  - σ^{passive} also contains the **viscous** stress, so this is just “active push = viscous drag + pressure”
- order is advected, rotated/stretched, and relaxed by the flow:

$$(\partial_t + \mathbf{u} \cdot \nabla)\mathbf{Q} - \mathbf{S}(\mathbf{W}, \mathbf{Q}) = \Gamma \mathbf{H}$$

- left term = **advection** of the order by the flow (carried along with the fluid)
- $\mathbf{S}(\mathbf{W}, \mathbf{Q})$ = the **co-rotational (generalized advection) term**: the flow **turns and tilts the rods**. When neighbouring fluid moves at different speeds (a velocity gradient $\mathbf{W} = \partial_i u_j$), it drags the alignment around – like sticks floating in a shearing current. *Note: this* $\mathbf{S}$ *is not the scalar order parameter* S.
- right term = **relaxation** toward equilibrium: $\mathbf{H} = -\delta F/\delta \mathbf{Q}$ = molecular field (the thermodynamic force from the free energy F, pushing $\mathbf{Q}$ to its minimum-F shape); Γ = rotational mobility (how fast it relaxes)
- **feedback loop**: order → active stress → flow → reorients order

Board 5: Generic instability and active turbulence

- the uniformly ordered state is **unstable** above a threshold:

$$\zeta_c \sim K/L^2$$

- extensile → **bend** instability; contractile → **splay** instability
- intrinsic active length scale: $\ell_{\text{active}} \sim \sqrt{K/\zeta}$ (vortex/defect spacing; shrinks as activity grows)
- above threshold: spontaneous $\pm 1/2$ defect pairs, chaotic flow = **active turbulence** (no inertia)

Board 6: Defects as active particles

- $+1/2$ defect self-propels: $v_{+1/2} \sim \zeta\, \ell_d/\eta$
- $-1/2$ defect: no net active force (three-fold symmetry)
- steady-state defect density: $n_{\text{defect}} \sim \zeta/K$
- **when defects matter**: above ζ_c (turbulence); transport and mixing; biology (stress and cell extrusion at $+1/2$ defects, morphogenesis)